JONATHAN TOLLEFSON

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ACADEMIC APPOINTMENTS

2025 - **University of Chicago**, Chicago, IL
Postdoctoral Fellow, Department of Sociology and Mansueto Institute for Urban Innovation

EDUCATION

2019 - 2025 **Brown University**, Providence, RI
Ph.D., Sociology

2016 - 2018 **University of Vermont**, Burlington, VT
M.S., Rubenstein School of Environment and Natural Resources

2011 - 2015 **Brown University**, Providence, RI
Sc.B. with honors, Social Analysis and Research

PUBLICATIONS

12. Demortain, D., Frickel, S., *Beck, S., *Gray, I., *Grundmann, R., *Hannud Abdo, A., *Kinniburgh, F., *Kitson, D., *Matzner, N., *Rea, C., ***Tollefson, J**., and *Wetts, R. (2026) "Computational science and technology studies? Assembling STS theory-methods packages with computation." Accepted for publication in *Science, Technology, & Human Values*.
*Authors listed alphabetically

11. **Tollefson, J.**, Frickel, S., Helgeson, J., and Gore, C. (2026) "Adapting to adaptation: Conceptual development and field formation in community resilience planning." Accepted for publication in *Ecology & Society*.

10. **Tollefson, J.** (2026) "Environment and the racialization of space in US cities." *Social Forces* (online first).

- IPUMS USA: Student Research Award (2023)
- American Sociological Association, Environmental Sociology Section: Graduate Student Paper Award - Honorable Mention (2024)
- Society for the Study of Social Problems, Environment and Technology Division: Brent K. Marshall Award (2024)
- Eastern Sociological Society: Candace Rogers Award - Honorable Mention (2024)
- Spatial Structures in the Social Sciences, Brown University: Graduate Student Paper Prize (2024)
- Featured in IBES spotlight article

9. **Tollefson, J.**, Frickel, S., Gore, C., and Helgeson, J. (2025) "Community resilience planning: What new methods reveal about the formation and transformation of a field." *WIREs Climate Change* 16(4): e70015.

8. Candipan, J. and **Tollefson, J.** (2024) "Machine learning and large-scale data for understanding urban inequality." Invited chapter for *The Oxford Handbook of the Sociology of Machine Learning* (Borch, C. and Pardo-Guerra, J.P., eds.).

7. **Tollefson, J.**, Frickel, S., Gonsalves, S., and Marlow, T. (2023) "Parks, people and pollution: A relational study of socioenvironmental succession." *City and Community* special issue, "Environmentalizing Urban Sociology" 22(4): 286-307.

6. **Tollefson, J.**, Frickel, S., Gonsalves, S., Marlow, T., Sucsy, R., Byrns, M., and Orpen-Tuz, M. (2023) "Early childcare and education in a post-industrial landscape: Inequalities in exposure to active and relic manufacturing in metropolitan Providence, Rhode Island." *Environmental Justice* 16(4): 309-320.

5. Frickel, S., and **Tollefson, J.** (2022) "When environmental inequality racialized: Historical evidence from Providence, Rhode Island." *Socius* 8: 1-14.

4. **Tollefson, J.**, Frickel, S, and Restrepo, M. (2021) "Feature extraction and machine learning techniques for identifying historic urban environmental hazards: New methods to locate lost fossil fuel infrastructure in US cities." *PLOS ONE* 16(8): e0255507.

- Finalist for the John Odland Student Paper Award, American Association of Geographers, Spatial Analysis and Modeling Group (2022)

3. **Tollefson, J.** and Panikkar, B. (2020) "Contested extractivism: Impact assessment, public engagement, and environmental knowledge production in Alaska's Yukon-Kuskokwim Delta." *Journal of Political Ecology* 27(1): 1166-1188.

- Graduate Student Paper Award, Rural Sociological Society, Natural Resources RIG (2021)

2. **Tollefson, J.** (2020) "Post-Fukushima discourse in the US press: Quantified knowledge, the technical object, and a panicked public." *Public Understanding of Science* 29(7): 670-687.

1. Panikkar, B. and **Tollefson, J.** (2018) "Land as material, knowledge, and relationships: Resource extraction and subsistence imaginaries in Bristol Bay, Alaska." *Social Studies of Science* 48(5): 715-739.

MANUSCRIPTS

Working Paper	**Tollefson, J.** "The environmental dimensions of HOLC redlining decisions." *Delivered at Population Association of America (Washington, DC. Apr. 10-13, 2025) and American Sociological Association (Chicago, IL. Aug. 8-12, 2025).*
Working Paper	**Tollefson, J.** "Industrial geography and urban inequality formation in the early 20th century."
Working Paper	With Frickel, S. "STS methods for historical environmental justice research." Invited chapter for *Common(s) in science and technology? Research-action in the SEEKCommons Network* (Murillo, L.F., Milia, M. and Bashir, I.B., eds.).
Working Paper	*Chaves, R., Tollefson, J., and *Long, M. "New methods for historical research on social and environmental geography in US cities." * Student co-authors

In Progress — Frickel, S., Demortain, D., Beck, S., Gray, I., Grundmann, R., Hannud Abdo, A., Kinniburgh, F., Kitson, D., Matzner, N., Rea, C., **Tollefson, J**., and Wetts, R. "Environmental knowledge and politics."

PUBLIC SCHOLARSHIP

Tollefson, J. (2023) "Redlining and segregation in Providence, RI." Essay for *Mapping Inequality: Redlining in New Deal America.* Richmond, VA: University of Richmond Digital Scholarship Lab.

- Media coverage: "Providence's Rent Stabilization Debate." March 12, 2026. *The College Hill Independent.*

Tollefson, J. and Frickel, S. (2021) "Gasworks, lost and found." *Urban Omnibus.* New York: The Architectural League of New York. July 1.

Tollefson, J. and Urso, M. (2016) "Black Lives Matter at St. Patrick's Day Parade." Op-ed in the *Providence Journal*. April 2.

DATASETS

Tollefson, J. (2026) Urban energy production in US cities. Supported by the SEEKCommons Project (National Science Foundation FAIROS RCN Award 2226425).

Tollefson, J (2026) Historical geographies of Chicagoland. Supported by CROCUS (Argonne National Laboratory) and Mansueto Institute for Urban Innovation (University of Chicago).

AWARDS

Brent K. Marshall Student Paper Award, Society for the Study of Social Problems, Environment and Technology Division (2024)

Candace Rogers Award, Eastern Sociological Society – Honorable Mention (2024)

Student Paper Award, American Sociological Association, Environmental Sociology Section – Honorable Mention (2024)

Paper Prize, Spatial Structures for the Social Sciences, Brown University (2024)

IPUMS USA Student Research Award (2023)

Finalist for the John Odland Student Paper Award, American Association of Geographers, Spatial Analysis and Modeling Group (2022)

Graduate Student Paper Award, Rural Sociological Society, Natural Resources RIG (2021)

First Place Poster Award, Dellinger Symposium, Superfund Research Program of the NIEHS (2021)

FELLOWSHIPS AND GRANTS

CROCUS research funding (Argonne National Laboratory). 2026.

- Research funding to develop new longitudinal data on urban built environments and industrial land uses in the Chicago area. Administered through CROCUS Lab in partnership with Mansueto Institute for Urban Innovation (University of Chicago).

National Science Foundation Socio-Environmental Knowledge Commons (SEEKCommons) Research Fellowship. 2025.

- Research fellowship to support the creation of public-facing data on historical urban built environments. Supported by NSF award #2226425

Interdisciplinary Opportunities Fellowship, *Institute at Brown for Environment and Society*. Full funding, 2024-25 academic year.

Research Fellow: Community Resilience and Risk. *Georgetown University / NIST-PREP*. Jan. – Dec. 2024.
- Georgetown University / National Institute of Standards and Technology

Joukowski Summer Research Award ($1,000), *Brown University*. 2024.

Asa and Horace Learned Fellowship, *Brown University.* Full funding, 2023-24 academic year.

National Institute of Standards and Technology Professional Research Experience Program (NIST-PREP) Fellowship. Full funding, Jan. – Jun. 2023.

Graduate Research, Training, and Travel Award ($3,145), *Institute at Brown for Environment and Society*. 2023.

Finalist (not awarded) – Raskin Scholarship, *Earth Science Information Partners*. 2022.

National Institute for Environmental Health Sciences, Superfund Research Program (NIEHS-SRP) Traineeship. Full funding, Jan. 2021 – Dec. 2021.
- Supported by NIEHS grant P42ES013660.

Collaborative Research Award ($2,000), *Brown University Department of Sociology*. 2021.

Collaborative Research Award ($2,000), *Brown University Department of Sociology*. 2020.

Graduate Fellowship, *Brown University.* Full funding, 2019-20 academic year.

Governance of Large Mine Permitting Research Fellowship. Full funding, 2016-17 and 2017-18 academic years.
- Supported by National Science Foundation award #1642226 (PI: Bindu Panikkar).

Florence Tinker Matteossian Memorial Grant ($4,214), *University of Vermont.* 2017.

Graduate Student Senate Mini-Grant, *University of Vermont.* 2017.

Rubenstein Graduate Student Association Mini-Grant, *University of Vermont*. 2017.

Karen T. Romer Undergraduate Teaching and Research Award ($3,000), *Brown University*. 2013.

Travel Grants

- 2024: Disaster Research Center Workshop Travel Grant, *University of Delaware Disaster Research Center*
- 2021: Dellinger Symposium Travel Award, *NIEHS Superfund Research Program*
- 2025, 2024: International conference travel grant, *Brown University*
- 2025, 2024, 2023, 2022, 2021: Graduate School Travel Grant, *Brown University*
- 2025, 2024, 2023, 2022, 2021: Sociology Dept. Travel Grant, *Brown University*
- 2022: GSC Travel Grant, *Brown University*

WORKSHOPS AND PANELS ORGANIZED

Historical and environmental foundations of spatial inequality. Session Organizer. *Population Association of America*. Seattle, WA. Mar. 31 - Apr. 3, 2027.

Historical and environmental foundations of urban inequality: Insights from new methodological assemblages. Panel Organizer (with K.A. Thomas). *Society for the Advancement of Socio-Economics.* Montreal, QC. Jul. 9-12, 2025. (Session canceled due to scheduling conflict).

Environmental Sociology Workshop. Organizer. *Brown University*. 2021 – 2025.

INVITED PRESENTATIONS

Tollefson, J. (2026). *Invited presentation*: Historical geographies of environmental segregation. *Social Science History Association.* Atlanta, GA. Nov. 19-22.

Tollefson, J. (2026). *Invited presentation.* Commoning and computation in environmental sociology: Advancing a historical perspective. *SEEKCommons Network Gathering.* University of California – Davis, Davis, CA. May 29.

Tollefson, J. (2024). *Keynote lecture.* What historic maps reveal about racial-environmental inequality in US cities. *Center for Urban and Regional Analysis Summer Speaker Series.* Ohio State University, Columbus, OH, and Online. Jun. 14.

Tollefson, J. (2024). *Invited presentation.* Environmental risk and the reorganization of urban inequality in the late 19th and early 20th century. *Spatial Structures in the Social Sciences, 2024 Graduate Fellow Lecture.* Providence, RI. Apr. 26.

Tollefson, J. (2023). *Invited presentation.* The development and transformation of community resilience planning research: Insights from domain-topic models with chained dimensions. *NIST Applied Economics Office Research Seminar Series*. Washington, DC. Nov. 3.

Tollefson, J. (2023). *Invited presentation.* The environmental dimensions of urban inequality, 1880-1930. *Spatial Data Science Symposium.* Online. Sept. 6.

Tollefson, J. (2022). *Invited presentation.* Energy geography and urban inequality: Computational techniques for historical research. *Computational Environmental STS Working Group (Laboratoire Interdisciplinaire Sciences Innovations Sociétés).* Paris, France. Jul. 10-13.

Tollefson, J. (2020). *Invited presentation.* Manufactured gas and historic environmental inequality. *Urban Theory and Data Lab (Harvard University).* Cambridge, MA. Oct. 29.

Tollefson, J. (2017). *Invited presentation.* Land use, power, and knowledge at the Northern resource frontier. *Rubenstein School of Environment and Natural Resources Graduate Symposium (University of Vermont).* Burlington, VT. Oct. 13.

CONFERENCE PRESENTATIONS

Tollefson, J. (2026). Environment, racialization, and territorial stigma: Evidence from HOLC texts. *Social Science History Association.* Atlanta, GA. Nov. 19-22.

Tollefson, J. (2026). Historicizing environmental sociology: Insights from science and technology studies. *American Sociological Association.* New York, NY. Aug. 7-11.

Tollefson, J. (2025). Nature, built environments, and risk in HOLC "redlining" decisions. *American Sociological Association.* Chicago, IL. Aug. 8-12.

Tollefson, J. (2025). Socio-environmental foundations of neighborhood segregation in US cities: New historical evidence. *Society for the Advancement of Socio-Economics.* Montreal, QC. Jul. 9-12. (Paper accepted, session canceled).

Tollefson, J. (2025). The environmental dimensions of HOLC "redlining" decisions. *Population Association of America*. Washington, DC. Apr. 10-13.

Tollefson, J. (2024). Environmental risk and the reorganization of urban inequality in the late 19th and early 20th centuries. *Society for the Study of Social Problems.* Montreal, QC. Aug. 9-11.

Tollefson, J. (2024). Siloed Interdisciplinarity?: A Multi-level Computational Text Analysis of Community Resilience Planning. *American Sociological Association.* Montreal, QC. Aug. 9-13.

Tollefson, J. (2024). Community resilience planning: What new methods reveal about the formation and transformation of a field. *Disaster Research Center 60th Anniversary Workshop.* Newark, DE. May 2-5.

Tollefson, J. (2024). Urban environmental inequality from 1880 to 1930. *Population Association of America.* Columbus, OH. Apr. 17-20.

Tollefson, J. (2024). The development and structure of community resilience planning research. *Eastern Sociological Society.* Washington, DC. Feb. 29 – Mar. 3.

Tollefson, J. (2023). Urban fossil fuel production and the changing scale of racial segregation, 1880-1930. *International Association of Public Health Scholars*. Baltimore, MD. Oct. 2-5.

Burns, A., and **Tollefson, J.** (2023). Inequality in the effects of pesticide exposure on infant health. *American Sociological Association.* Philadelphia, PA. Aug. 17-21.

Tollefson, J. (2023). How industry segregated cities: Using computational data extraction to reexamine the formation of early-20th century segregation. *American Sociological Association.* Philadelphia, PA. Aug. 17-21. (Paper accepted, could not attend due to scheduling conflict).

Burns, A., and **Tollefson, J.** (2023). Inequality in the effects of pesticide exposure on infant health. *Population Association of America*. New Orleans, LA. Apr. 12-15.

Tollefson, J. (2023). Environmental inequality and the changing scale of urban segregation in the early 20th century. *Pacific Sociological Association.* Bellevue, WA, Mar. 30 – Apr. 2.

Tollefson, J. (2023). What was the role of environmental risk in the early formation of neighborhood-level segregation? *American Association of Geographers.* Denver, CO. Mar. 23-27.

Tollefson, J. and Silberstein, S. (2023). Right-wing extremism online: A quantitative and critical discourse analysis of the Parler website. *American Association for Applied Linguistics.* Portland, OR. Mar. 18-21.

Tollefson, J. (2022). Brownfields and green spaces: Neighborhood change and urban park access in a post-industrial city. *American Sociological Association.* Los Angeles, CA. Aug. 5-9.

Frickel, S., and **Tollefson, J.** (2022). When environmental inequality racialized: Evidence from a historical case study. *American Sociological Association.* Los Angeles, CA. Aug. 5-9.

Tollefson, J. (2022). Agricultural land conversion and new geographies of suburban environmental risk. *Pacific Sociological Association.* Sacramento, CA. Apr. 7-9.

Tollefson, J. (2022). Gastown: Fossil fuels and the changing nature of urban environmental inequality at the turn of the 20th century. *Eastern Sociological Society.* Boston, MA. Mar. 10-13.

Tollefson, J. (2022). New methods to locate historic fossil fuel infrastructures: How computational techniques can illuminate 19th century urban environmental hazards. *American Association of Geographers.* New York, NY and online. Feb. 25 – Mar 1.

- Presented as part of John Odland Award session.

Tollefson, J. (2021). Urban environmental inequality in the 19th century and the toxic legacy of the manufactured gas industry: Unearthing early fossil fuel infrastructure in American cities (poster). *Dellinger Symposium of the Superfund Research Program.* Apr. 28.

- Superfund Research Program of the NIEHS, Dellinger Symposium: First Place Poster Award

Tollefson, J. (2021). Urban environmental inequality and the toxic legacy of the manufactured gas industry: Unearthing 19th century fossil fuel infrastructure in American cities. *Pacific Sociological Association.* Online. Mar. 17-21.

Tollefson, J. (2018). Mining, public engagement, and contentious land imaginaries in Bristol Bay and the Yukon-Kuskokwim Delta, Alaska. *American Association of Geographers.* New Orleans, LA. Apr. 10-14.

Tollefson, J. (2017). Knowledge, sovereignty, and resource extraction in rural Alaska: A case study of the Donlin Gold public comment process. *ArcticNet Annual Scientific Meeting.* Quebec City, QC. Dec. 11-15.

Tollefson, J. (2017). Land as material, habitat and survival: Resource materiality and mining in Bristol Bay and the Yukon-Kuskokwim Delta, Alaska. *Society for Social Studies of Science.* Boston, MA. Aug. 30 – Sept. 2.

Tollefson, J. (2016). Resource materiality and resistance to mineral development in Bristol Bay, Alaska. *ArcticNet Annual Scientific Meeting.* Winnipeg, MB. Dec. 5-9.

TEACHING

UNIVERSITY TEACHING

Brown University, Environmental Studies / "Environmental justice and the city" / *Instructor of Record*. 2025.

I.E. Business School and Brown University School of Professional Studies / "Society and Culture in the Age of Globalization" / *Teaching Associate* / Prof. Meghan Kallman. 2025, 2024, 2023, 2022.

Brown University, Sociology / "Organizations and Society" / *Teaching Assistantship* / Prof. Rachel Wetts. 2022.

Brown University, Sociology / "Methods for Interdisciplinary Environmental Research" / *Teaching Assistantship* / Prof. Rachel Wetts. 2020.

University of Vermont, Environmental Studies / "Human Ecology, Health, and Sustainability in the Circumpolar Arctic" / Co-taught with Prof. Bindu Panikkar. 2018.

University of Vermont, Rubenstein School of Environment and Natural Resources / "Social Processes and the Environment" / *Teaching Assistantship* / Prof. Cecilia Danks. 2017.

University of Vermont, Environmental Studies / "Environmental Policy and Activism" / *Teaching Assistantship* / Prof. Trish O'Kane. 2017.

University of Vermont, Rubenstein School of Environment and Natural Resources / "Race, Culture, and Natural Resources" / *Teaching Assistantship* / Prof. Clare Ginger. 2017.

Guest lecturing

Brown University, Sociology / "The Geography of Urban Inequality" / *Guest Lecturer* / Prof. Jennifer Candipan. 2024.

Brown University, Environmental Studies / "Environmental Studies of Pollution" / *Guest Lecturer* / Profs. Meredith Hastings and Scott Frickel. 2026, 2022, 2021.

COMMUNITY TEACHING

Extraordinary Youth Ensemble (Providence, RI) / *Administrator and teacher*. 2015-2016.

- Developed curriculum and successfully obtained biannual grants from the Rhode Island State Council for the Arts to provide instrument and music history instruction to Providence, RI area high school students.

Trinity Academy for the Performing Arts (Providence, RI) / *Visiting Artist, Music Instruction*. 2014 –2015.

Breakthrough Providence (Providence, RI) / *Teacher, "Demonstration and Organizing."* 2014 – 2015.

- Developed original student-centered curriculum for 7th-9th grade students. Taught weekly lessons and advised student projects on the role of organizing in bringing about social and political change. Advised teaching assistant.

SERVICE

Society for the Study of Social Problems Environment & Tech. Division award committee. 2025, 2026.

Graduate Student Council (Brown University), Department of Sociology representative. 2021 – 2025.

Presider, "Urban Environments" roundtable session, *American Sociological Association* annual meeting. Montreal, QC. Aug. 9-13, 2024.

Poster Judge, *International Association of Public Health Scholars* annual meeting. Baltimore, MD. Oct. 2-5, 2023.

Presider, "Urban Sociology" panel session, *American Sociological Association* annual meeting. Philadelphia, PA. Aug. 17-21, 2023. (Could not attend due to scheduling conflict).

Presider, "Environmental Justice" panel session, *Pacific Sociological Association* annual meeting. Bellevue, WA. Mar. 30 – Apr. 2, 2023.

Rural Sociological Society Natural Resources Research Interest Group award committee. 2022.

"Getting to Know Brown and Rhode Island," panel presentation for incoming graduate students. 2022.

Reviewer

American Sociological Review; Social Forces; City & Community; Public Understanding of Science; Population and Environment; The Extractive Industries and Society; Journal of Political Ecology; Journal of Computational Social Science; PLOS ONE.

AFFILIATIONS

INSTITUTES, LABS, AND WORKING GROUPS

Mansueto Institute for Urban Innovation (University of Chicago). 2025 – Present.

Computational Environmental STS Working Group (Laboratoire Interdisciplinaire Sciences Innovations Sociétés Paris; Technical University Munich; University of Nottingham; Johns Hopkins University; University of Lisbon; Brown University; University of Chicago). 2022 – Present.

Institute at Brown for Environment and Society (Brown University). 2020 – 2025.

Urban Theory and Data Lab (Harvard University). 2020 – 2022.

Gund Institute for Environment (University of Vermont). 2017 – 2018.

PROFESSIONAL MEMBERSHIPS

American Sociological Association

- Section Memberships: Environmental Sociology; Community and Urban Sociology; Science, Knowledge, and Technology

American Association of Geographers
Eastern Sociological Society
Interdisciplinary Association for Population Health Science
Pacific Sociological Association
Population Association of America
Rural Sociological Society
Society for the Study of Social Problems

LANGUAGE AND TECHNICAL

Language: Japanese (rusty)

*I use an array of computational tools in my research and teaching, including statistical analysis in **Python, R**, and **Stata**; spatial analysis in **Python, R**, **ArcGIS**, and **QGIS**; automation, web scraping, text analysis, and machine learning applications in **Python** and **R**; mixed qualitative-quantitative analysis using the **CorText** and **Sashimi** methodologies; and qualitative analysis in **NVivo** and **HyperResearch**.*